

ROBUST TRANSFER LEARNING WITH UNRELIABLE SOURCE DATA

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This paper addresses challenges in robust transfer learning stemming from ambiguity in Bayes classifiers and weak transferable signals between the target and source distributions. We introduce a novel quantity called the “ambiguity level” that measures the discrepancy between the target and source regression functions, propose a simple transfer learning procedure, and establish a general theorem that shows how this new quantity is related to the transferability of learning in terms of risk improvements. Our proposed “Transfer Around Boundary” (TAB) method, with a threshold that balances the performance contributions of the target and source data, is shown to be both efficient and robust, improving classification while avoiding negative transfer. Moreover, we demonstrate the effectiveness of the TAB model on nonparametric classification and logistic regression tasks, achieving upper bounds which are optimal up to logarithmic factors. Simulation studies lend further support to the effectiveness of TAB. We also provide simple approaches to bound the excess misclassification error without the need for specialized knowledge in transfer learning.

1. Introduction. Previous experiences can offer valuable information for learning new tasks. Human learners often transfer their existing knowledge gained from previous tasks to new and related ones. *Transfer learning* refers to statistical learning tasks where a portion of the training data is generated from a similar but nonidentical distribution to the data distribution for which we seek to make inferences. The objective is then to transfer knowledge from such source data to improve learning in the related target task. Such problems, where there is a divergence between the data-generating distributions, arise in many real application problems, including computer vision (Li et al. (2020), Tzeng et al. (2017)), natural language processing (Ruder et al. (2019), Wang and Zheng (2015)), speech recognition (Huang et al. (2013)) and genre classification (Choi et al. (2017)). See Storkey (2008), Pan and Yang (2010), and Weiss, Khoshgoftaar and Wang (2016) for an overview. Also, similar problems have been studied by both statisticians and many other communities under different names, including label noise (Frenay and Verleysen (2014), Scott, Blanchard and Handy (2013), Cannings, Fan and Samworth (2020), Blanchard et al. (2021), Reeve and Kaban (2019), Scott and Zhang (2019)), domain adaptation (Scott (2019), Ben-David et al. (2010a,b), Mansour, Mohri and Rostamizadeh (2009a)), multitask learning (Caruana (1997), Maurer, Pontil and Romera-Paredes (2016)) or distributional robustness (Sinha, Namkoong and Duchi (2018), Christiansen et al. (2020)).

We focus here on the transfer learning setting in the context of binary classification since it is not only fundamental in statistical learning and has been extensively investigated in diverse contexts, but also because it provides a framework that is particularly conducive to algorithms that seek to exploit relationships between the target and source distributions. For transfer learning theory of linear regression models, see Chen, Owen and Shi (2015), Bastani

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(2021) under the setting of finite covariate dimensions or Gross and Tibshirani (2016), Ollier and Viallon (2017), Li, Cai and Li (2022) under the high-dimensional regime with lasso-based penalties. See also Tian and Feng (2023) for generalized linear models and Cai and Pu (2024) for nonparametric regression.

To set up the framework, suppose that a labeled data sample (relatively *small* in size, typically) is drawn from Q , the target distribution we wish to make statistical inferences about. Also, let P be the source distribution from which we wish to transfer knowledge. We suppose a labeled data sample (relatively *large* in size, typically) is drawn from P . The corresponding random pairs of Q and P distributions are denoted by (X, Y) and (X^P, Y^P) on $\mathbb{R}^d \times \{0, 1\}$, respectively. Let $\eta^Q, \eta^P : \mathbb{R}^d \rightarrow [0, 1]$ denote the target and source regression functions, that is,

$$\eta^Q(x) = Q(Y = 1 \mid X = x) \quad \text{and} \quad \eta^P(x) = P(Y^P = 1 \mid X^P = x).$$

The key question is how much information or knowledge can be transferred from P to Q given observations from both distributions. Note that the Bayes classifier $f_Q^*(\cdot) = \mathbf{1}\{\eta^Q(\cdot) \geq \frac{1}{2}\}$ minimizes the misclassification rate $Q(Y \neq f(X))$ over all classifiers. Therefore, we define the excess risk of any empirical classifier $\hat{f}$ as

$$\mathcal{E}_Q(\hat{f}) = Q(Y \neq \hat{f}(X)) - Q(Y \neq f_Q^*(X)).$$

The aforementioned key question leads us to the task of constructing an empirical classifier that accelerates the convergence of excess risk to zero by utilizing labeled data samples from *both* Q and P .

1.1. Related literature. Recent research has aimed to bridge the gap between limited theoretical understanding and significant practical achievements in transfer classification learning. To set up the problem, it is necessary to make some assumptions about the similarity between the target and source distributions, which are both useful for theory and practical implementation. Various approaches have been proposed and explored in the literature to measure this similarity, including divergence bounds, covariate shift and label shift. Some methods focus on test error bounds that rely on measures of discrepancy between Q and P , such as modified total-variation or Rényi divergence, between the target and source distributions (Ben-David et al. (2010a,b), Mansour, Mohri and Rostamizadeh (2009a,b), Germain et al. (2013), Cortes, Mohri and Muñoz Medina (2019)). This line of work produces distribution-free risk rates, primarily expressed in terms of n_P alone, but the obtained rates do not converge to zero with increasing sample size. In other words, these frameworks cannot produce a faster convergence of excess risk if their proposed divergences are nonnegligible. Nonetheless, consistent classification is proved achievable regardless of a nonnegligible divergence even when $n_Q = 0$, provided that certain additional structures on the target and source distributions are present (Ben-David et al. (2010b)).

Two common additional structures include covariate shift and label shift (or posterior shift). Covariate shift (Gretton et al. (2009), Quionero-Candela et al. (2009), Kpotufe and Martinet (2021)) considers scenarios where the conditional distributions of the response given the covariates are identical across Q and P or $\eta^Q = \eta^P$, but marginal distributions of covariates are different. Label shift, on the other hand, assumes an identical or similar target and source marginal distributions Q_X and P_X , but the conditional probabilities η^Q and η^P differ.

Most previous work in label shift can be divided into two branches. On the one hand, some frameworks do not require identical Bayes classifiers, that is, $(\eta^P - 1/2)(\eta^Q - 1/2) \geq 0$, but impose very specific relations between η^Q and η^P , such as the literature on *label noise* (Frenay and Verleysen (2014), Scott, Blanchard and Handy (2013), Cannings, Fan and Samworth (2020), Blanchard et al. (2021), Reeve and Kaban (2019), Natarajan et al. (2017),

Scott and Zhang (2019)). This literature typically considers the labels of the source domain as “corrupted” in a certain way compared to the true labels in the target distribution, with the corrupted labels usually coupled with the true labels given the same feature data X . For instance, a common assumption (Reeve and Kaban (2019), Natarajan et al. (2017)) is that the $Y^P \mid X^P = x$ is equal to $Y \mid X = x$ up to a constant probability of label flipping, that is, in our terminology

$$P(Y^P = 1 \mid Y = 0, X^P = X = x) = \pi_0 \quad \text{and} \quad P(Y^P = 0 \mid Y = 1, X^P = X = x) = \pi_1,$$

for some constants $\pi_0, \pi_1 \in (0, 1)$. Under this setting, $\eta^P = (1 - \pi_0 - \pi_1)\eta^Q + \pi_0$, which is linear in η^Q . Given knowledge of the specific form of η^P , it is feasible to modify learning algorithms to efficiently infer the target Bayes classifier. In another special type of label shift problem, Maity, Sun and Banerjee (2022) assume that $P_{X|Y} = Q_{X|Y}$, which is convenient for estimating the joint distribution of (X, Y) . However, all their proposed estimators are tailored to fit specific assumptions and may not be applicable to more general relations between η^Q and η^P . On the other hand, recent work such as Cai and Wei (2021) and Hanneke and Kpotufe (2019) have introduced more general label shift settings that impose relatively mild and general conditions on the relation between η^P and η^Q , in addition to assuming identical Bayes classifiers. Cai and Wei (2021) require a lower-bounded signal strength of η^P relative to η^Q , besides the assumption of identical Bayes classifiers, that is,

$$\left(\eta^P - \frac{1}{2}\right)\left(\eta^Q - \frac{1}{2}\right) > 0, \quad \left|\eta^P - \frac{1}{2}\right| \geq C_\gamma \left|\eta^Q - \frac{1}{2}\right|^\gamma,$$

for some positive γ and C_γ , and derive a faster risk convergence rate with the transfer exponent γ .

The aforementioned approaches have limitations in that they rely on specific and often untestable assumptions. More importantly, they may be ineffective in cases where both types of ambiguity in the source data are present; that is, when there is no strong relationship between η^Q and η^P , and discrepancies persist between the Bayes classifiers of the target and source domains. The only other work of which we are aware that allows such general ambiguous source data is Reeve, Cannings and Samworth (2021). Reeve, Cannings and Samworth (2021) assume that η^P can be well approximated by a set of regression functions $g_1(\eta^Q), \dots, g_{L^*}(\eta^Q)$ that are no less informative than a linear transformation of η^Q .

1.2. Main contribution. In contrast to existing work, this paper considers a scenario where the Bayes classifiers can arbitrarily differ without imposing further conditions on the source and target distributions. We now introduce an important quantity for capturing the relative information transferred from the source data, which we refer to as the signal strength:

$$s(x) := \begin{cases} \left|\eta^P(x) - \frac{1}{2}\right|, & \text{sgn}\left(\eta^Q(x) - \frac{1}{2}\right) \times \left(\eta^P(x) - \frac{1}{2}\right) \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

Our main assumption involves measuring the ambiguity level of the source data based on the expected signal strength around the classification boundary $\{x : |\eta^Q(x) - \frac{1}{2}| \leq z\}$ for small z :

$$\mathbb{E}_{(X,Y) \sim Q} \left[\left| \eta^Q(X) - \frac{1}{2} \right| \mathbf{1} \left\{ s(X) \leq C_\gamma \left| \eta^Q(X) - \frac{1}{2} \right|^\gamma, \left| \eta^Q(X) - \frac{1}{2} \right| \leq z \right\} \right] \leq \varepsilon(z),$$

given some constants $\gamma, C_\gamma > 0$. This quantity captures the inherent risk posed by hard-to-classify boundary points—those for which η^Q is close to $1/2$ and η^P provides a weak signal relative to η^Q . The explicit form of $\varepsilon(z)$ is provided for some special examples in Section 2.3. If $s(x) \geq C_\gamma |\eta^Q(x) - 1/2|^\gamma$ covers all of Ω , then $\varepsilon(z) = 0$ and the setting reduces to that of Cai and Wei (2021) with a strong relative signal of η^P across the entire feature space Ω .

Given the measure above, we propose a simple but effective classifier that can surprisingly adapt to any level of ambiguity in the signal conveyed by η^P , named the *Transfer Around Boundary* (TAB) classifier, or the TAB model:

$$\hat{f}_{\text{TAB}}(x) = \begin{cases} \mathbf{1}\left\{\hat{\eta}^Q(x) \geq \frac{1}{2}\right\}, & \text{if } \left|\hat{\eta}^Q(x) - \frac{1}{2}\right| \geq \tau, \\ \hat{f}^P(x), & \text{otherwise,} \end{cases}$$

where $\hat{\eta}^Q$ is an estimate of η^Q obtained by the target data and $\hat{f}^P$ is a classifier obtained by the source data. To clarify our decision-making process, we rely on $\hat{\eta}^Q$ as the final prediction when it deviates significantly from $1/2$. Otherwise, we switch to the prediction made by the source data. Our TAB classifier is easy to implement and does not require careful classifier design.

In Section 3.1, we present two related general convergence theorems to showcase this classifier, with a proper choice of $\tau > 0$, utilizing both the relative signal of η^P and the ambiguity level $\varepsilon(z)$ from the source data. The target data involved in this classifier is noteworthy for its dual role: not only does it preserve a classical upper bound on the excess risk if the source data is unreliable, but it also helps to rule out the source data ambiguity when η^Q is far from $1/2$, considering only necessary ambiguity in the risk bounds.

We then apply our general convergence results to nonparametric classification as well as logistic regression to illustrate the potential of our transfer learning method on parametric classification. The results for these two important cases show that our method is indeed nearly minimax optimal and our measure of the ambiguity level is effective, extending results in previous literature.

Nonparametric classification. Suppose that η^Q is β -smooth (Condition 2), margin assumption holds for η^Q with parameter α (Assumption 1), and strong density condition (Condition 3) holds for Q_X and P_X . Let Π^{NP} denote the set of all such distribution pairs (Q, P) .

At the moment, we consider the scenario where η^P is sufficiently smooth. Suppose that Π_S^{NP} is a subset of Π^{NP} such that η^P is β_P -smooth with $\beta_P \geq \gamma\beta$. Then we show that the minimax excess risk satisfies

$$\begin{aligned} & \left(n_P^{-\frac{\beta(1+\alpha)}{2\gamma\beta+d}} + \varepsilon(cn_Q^{-\frac{\beta}{2\beta+d}})\right) \wedge n_Q^{-\frac{\beta(1+\alpha)}{2\beta+d}} \\ & \lesssim \inf_{\hat{f}} \sup_{(Q,P) \in \Pi_S^{\text{NP}}} \mathbb{E} \mathcal{E}_Q(\hat{f}) \\ & \lesssim \left(n_P^{-\frac{\beta(1+\alpha)}{2\gamma\beta+d}} + \varepsilon(c \log(n_Q \vee n_P) n_Q^{-\frac{\beta}{2\beta+d}})\right) \wedge (\log^{1+\alpha}(n_Q \vee n_P) n_Q^{-\frac{\beta(1+\alpha)}{2\beta+d}}), \end{aligned}$$

for some constant $c > 0$ that is independent of α .

With the optimal choice of $\tau \asymp \log(n_Q \vee n_P) n_Q^{-\frac{\beta}{2\beta+d}}$, the upper bound we obtain using the TAB classifier with K -NN classifier components is optimal up to some logarithmic terms of $n_Q \vee n_P$. This indicates that a necessary and sufficient condition for the source data to improve the excess risk rate is a large source data sample size such that $n_P \gg n_Q^{\frac{2\gamma\beta+d}{2\beta+d}}$, paired with a small ambiguity level $\varepsilon(z) \ll z^{1+\alpha}$.

Additionally, we provide the optimal rate considering the “band-like” ambiguity condition between η^Q and η^P . A particularly interesting example is to assume that $\sup_{x \in \Omega} |\eta^P(x) - \eta^Q(x)| \leq \Delta$, as it allows for effective transfer learning by ensuring there are only slight deviations across domains. In this case, the minimax optimal excess risk becomes $(n_P^{-\frac{\beta(1+\alpha)}{2\beta+d}} + \Delta^{1+\alpha}) \wedge n_Q^{-\frac{\beta(1+\alpha)}{2\beta+d}}$, up to some logarithmic terms of $n_Q \vee n_P$. Note that we do not require any smoothness condition on η^P in the band-like ambiguity case.

Logistic regression. If the covariates follow the standard normal random design (see (22) for a precise definition), and the target and source logistic regression coefficient pair (β_Q, β_P) belongs to

$$\Theta(s, \Delta) = \{(\beta_Q, \beta_P) : \|\beta_Q\|_0 \leq s, \angle(\beta_Q, \beta_P) \leq \Delta\},$$

for some $0 \leq \Delta \leq \pi/2$, where $\angle(\beta_Q, \beta_P)$ denotes the angle between directions β_Q and β_P , we show that the minimax optimal excess risk satisfies

$$\begin{aligned} \left(\frac{s \log d}{n_P} + \Delta^2\right) \wedge \frac{s \log d}{n_Q} &\lesssim \inf_{\hat{f}} \sup_{(Q, P) \in \Pi^{\text{LR}}} \mathbb{E} \mathcal{E}_Q(\hat{f}) \\ &\lesssim \left(\frac{s \log d}{n_P} + \Delta^2\right) \wedge \left(\frac{s \log d}{n_Q} \log^2(n_Q \vee n_P)\right). \end{aligned}$$

Thus, when $n_P \gg n_Q$ and $\Delta \ll \frac{s \log d}{n_Q}$, the source data improves the convergence rate of the excess risk. Importantly, we only assume a “small cone condition” between β_Q and β_P , in contrast to the “small contrast condition” in previous works (Li, Cai and Li (2022), Tian and Feng (2023)), that is, $\|\beta_Q - \beta_P\|_q$ is small for some $q \in [0, 1]$. In addition, unlike the small contrast condition, which implicitly assumes sparsity patterns of β_P through l_q norms with $q \leq 1$, our constructed parametric space does not impose any sparsity conditions on β_P . In the setting without access to source data, our upper bound $\frac{s \log d}{n_Q}$ is tighter than the $(\frac{s \log d}{n_Q})^{\frac{2}{3}}$ bound obtained in Theorem 7 of Abramovich and Grinshtein (2019), assuming the same margin parameter $\alpha = 1$.

As demonstrated by the two examples above, the TAB classifier preserves the performance of the target data in the presence of an unreliable source and even enhances it when the source dataset is sufficiently large and ambiguity is low, that is, $n_P \gg n_Q$ and $\Delta \ll \sqrt{\frac{s \log d}{n_Q}}$.

While the target excess risk provided by the target data, $\mathcal{E}_Q(\hat{f}^Q)$, has been extensively studied in the literature, much less is known about the target excess risk $\mathcal{E}_Q(\hat{f}^P)$ provided by the source data, which is one key component in deducing the excess risk bound. To fill this gap and gain better insight into the contribution of the source data, we present a general result in Section 3.2 that provides a direct upper bound on $\mathcal{E}_Q(\hat{f}^P)$ in terms of $\mathcal{E}_P(\hat{f}^P)$, which can be obtained through conventional theoretical analysis. Therefore, it is feasible to bound the excess risk using conventional statistical learning tools, without requiring specialized expertise in transfer learning. By providing such a general and accessible theoretical framework, our approach has the potential to make transfer learning more accessible.

1.3. Notation and organization. We introduce some notation to be used throughout the paper. For any $q \in [0, \infty]$ and vector $x = (x_1, \dots, x_d) \in \mathbb{R}^d$, we write $\|x\|_q$ for its l_q norm. We write $\|x\|$ or $\|x\|_2$ for the Euclidean norm of x , and, given $r > 0$, we write $B(x, r)$ or $B_d(x, r)$ for the closed Euclidean ball of radius r centered at x . For two probability measures μ, ν on any general space, if μ is absolutely continuous with respect to ν , we write $\frac{d\mu}{d\nu}$ for the Radon–Nikodym derivative of μ with respect to ν . Write λ as the Lebesgue measure on $\mathbb{R}^d$. Let $a \wedge b$ and $a \vee b$ denote the minimum and maximum of a and b , respectively. Let $a_n \lesssim b_n$ denote $|a_n| \leq c|b_n|$ for some constant $c > 0$ when n is large enough. Let $a_n \gtrsim b_n$ denote $|a_n| \geq c|b_n|$ for some constant $c > 0$ when n is large enough. Let $a_n \asymp b_n$ denote $1/c \leq |a_n|/|b_n| \leq c$ for some constant $c > 1$ when n is large enough. Let $a_n \xrightarrow{n \rightarrow \infty} \infty$ denote that a_n tends to infinity with n growing to infinity. Let $a_n \ll b_n$ denote $|a_n|/|b_n| \rightarrow 0$ when n is large enough. Let $a_n \gg b_n$ denote $|b_n|/|a_n| \rightarrow 0$ when n is large enough. Let $[a]$ be the maximum integer that is less equal than a for any real value a . Finally, we assume $0^0 = 0$ for simplicity.

We state our main working assumptions and measure of ambiguity, called the *ambiguity level*, in Section 2. We provide two general convergence results on the risk of transfer learning in Section 3.1. Section 3.2 presents an approach to bound the signal transfer risk, a crucial part of general convergence results, in terms of the excess risk rate studied in the conventional statistical learning literature. In Sections 4 and 5, we apply our results to nonparametric classification and logistic regression, respectively, and provide upper and lower bounds on the excess risk. In Section 6, we present simulation results for nonparametric classification and logistic regression, supporting the theoretical properties of our proposed method.

2. Model.

2.1. Problem formulation. For two Borel-measurable distributions P and Q , both taking values in $\mathbb{R}^d \times \{0, 1\}$, we observe two independent random samples, the *source* data $\mathcal{D}_P = \{(X_1^P, Y_1^P), \dots, (X_{n_P}^P, Y_{n_P}^P)\} \stackrel{\text{i.i.d.}}{\sim} P$ and the *target* data $\mathcal{D}_Q = \{(X_1, Y_1), \dots, (X_{n_Q}, Y_{n_Q})\} \stackrel{\text{i.i.d.}}{\sim} Q$. Suppose that $n_P \xrightarrow{n_Q \rightarrow \infty} \infty$. Our goal is to improve the target data empirical classifier by transferring useful information from the source data. Consider the marginal probability distributions of X for the P and Q distributions, denoted by P_X and Q_X , respectively. Let $\Omega_P := \text{supp}(P_X)$ and $\Omega := \text{supp}(Q_X)$ represent the support sets of P_X and Q_X . The regression functions for the source and target distributions are respectively defined as follows:

$$(1) \quad \begin{aligned} \eta^P(x) &= \begin{cases} P(Y^P = 1 \mid X^P = x), & x \in \Omega_P, \\ 1/2, & \text{otherwise,} \end{cases} \\ \eta^Q(x) &= \begin{cases} Q(Y^P = 1 \mid X^P = x), & x \in \Omega, \\ 1/2, & \text{otherwise.} \end{cases} \end{aligned}$$

The goal of a classification model is to forecast the label Y based on the value X . The effectiveness of a decision rule $f : \mathbb{R}^d \rightarrow \{0, 1\}$ is evaluated by its misclassification rate with respect to the target distribution, which is defined as follows:

$$R(f) := Q(Y \neq f(X)).$$

The Bayes classifier (or Bayes estimator, Bayes decision rule) $f_Q^*(x) = \mathbf{1}\{\eta^Q(x) \geq 1/2\}$ is the minimizer of $R(f)$ over all Borel functions defined on $\mathbb{R}^d$ and taking values in $\{0, 1\}$. We similarly define the Bayes decision rule for η^P as $f_P^*(x) = \mathbf{1}\{\eta^P(x) \geq 1/2\}$. Since the Bayes decision rule $f_Q^*(x)$ is a minimizer of the misclassification rate $R(f)$, the performance of any empirical classifier $\hat{f} : \mathbb{R}^d \rightarrow \{0, 1\}$ can be then measured by the excess risk (on the target distribution):

$$(2) \quad \mathcal{E}_Q(\hat{f}) = R(\hat{f}) - R(f_Q^*) = 2\mathbb{E}_{(X,Y) \sim Q} \left[\left| \eta_Q(X) - \frac{1}{2} \right| \mathbf{1}\{\hat{f}(X) \neq f_Q^*(X)\} \right].$$

The last equality is the dual representation of the excess risk (Györfi (1978)). Given the excess risk defined in (2), the objective of transferring useful information from the source data can be reformulated as the task of constructing an empirical classifier that improves the excess risk, which is accomplished by utilizing labeled data samples drawn from both the target and source distributions.

The rate at which the excess risk $\mathcal{E}_Q(\hat{f})$ converges to zero depends on the assumptions made about the target and source distributions (Q, P). In classification, a common assumption is the margin assumption (Audibert and Tsybakov (2007), Mammen and Tsybakov (2004)). This assumption is used to measure the behavior of Q_X with respect to the distance between $\eta^Q(X)$ and $1/2$, which is essential for determining the convergence rate of the excess risk.

ASSUMPTION 1 (Margin). There exists some constant $\alpha \geq 0$, $C_\alpha > 0$ such that for any $t > 0$, we have $\mathbb{Q}_X(0 < |\eta^Q(X) - 1/2| \leq t) \leq C_\alpha t^\alpha$.

Instead of assuming identical Bayes classifiers for η^Q and η^P over $x \in \Omega$ as in some existing literature (Hanneke and Kpotufe (2019), Cai and Wei (2021)), we allow them to differ. This is a natural and necessary relaxation because, in almost every real-world application, although the target and source domains may exhibit high similarity, they usually still have slightly different decision boundaries. Therefore, it is crucial to assess the impact of an unreliable source distribution on the optimal rates of classification.

2.2. *Source data ambiguity.* In this subsection, we provide a detailed discussion of the condition that characterizes the ambiguity of an unreliable source distribution. Here, we introduce the signal strength function, which measures the relative signal of η^P compared to η^Q . This function is crucial in capturing the efficacy of the source data for the classification task under Q .

DEFINITION 1 (Signal strength). The signal strength of η^P relative to η^Q is defined as

$$\begin{aligned} s(x) &:= \left\{ \operatorname{sgn}\left(\eta^Q(x) - \frac{1}{2}\right) \times \left(\eta^P(x) - \frac{1}{2}\right) \right\} \vee 0 \\ &= \begin{cases} \left|\eta^P(x) - \frac{1}{2}\right|, & \operatorname{sgn}\left(\eta^Q(x) - \frac{1}{2}\right) \times \left(\eta^P(x) - \frac{1}{2}\right) \geq 0, \\ 0, & \text{otherwise,} \end{cases} \end{aligned}$$

for any $x \in \Omega$.

It is reasonable to consider the signal strength as nonzero only when $(\eta^P(x) - 1/2) \times (\eta^Q(x) - 1/2) \geq 0$, indicating that the target and source data provide consistent information about the Bayes classification boundary. In this case, the source data $\eta^P(x)$ is beneficial for classifying the target data, and the signal strength is measured by $|\eta^P(x) - 1/2|$. Conversely, when $(\eta^P(x) - 1/2)(\eta^Q(x) - 1/2) < 0$, the source data does not provide useful information for classifying x in the target data, and the signal strength at x is zero.

Next, we present the main working assumption that measures the ambiguity level of the source data, based on the signal strength.

ASSUMPTION 2 (Ambiguity Level). For some given γ , $C_\gamma > 0$, the following condition holds for a specific continuous function $\varepsilon(z; \gamma, C_\gamma)$ that is monotone increasing with $z \in [0, 1/2]$:

$$(3) \quad \mathbb{E}_{(X,Y) \sim Q} \left[\left| \eta^Q(X) - \frac{1}{2} \right| \mathbf{1} \left\{ s(X) \leq C_\gamma \left| \eta^Q(X) - \frac{1}{2} \right|^\gamma, \left| \eta^Q(X) - \frac{1}{2} \right| \leq z \right\} \right] \leq \varepsilon(z; \gamma, C_\gamma).$$

We abbreviate $\varepsilon(z; \gamma, C_\gamma)$ as $\varepsilon(z)$ when there is no need to specify γ and C_γ .

The expression in the expectation operator of (3) can be divided into two factors. The first factor represents the distance between $\eta^Q(X)$ and $1/2$, which corresponds to the dual representation of the excess risk (2). The second factor is crucial for understanding transfer learning ambiguity. On the one hand, the constraint $s(X) \leq C_\gamma |\eta^Q(X) - 1/2|^\gamma$ indicates a lack of strong signal from the source data relative to the target data. On the other hand, the constraint $|\eta^Q(X) - 1/2| \leq z$ describes the hard-to-classify boundary points of the target distribution. The target data in this region cannot significantly aid in classification. Therefore, our indicator precisely captures the challenging boundary region where classification becomes difficult

using either the target or source data. As will be seen later, only the behavior around $z = 0$ matters for the asymptotics of transfer learning.

The ambiguity level $\varepsilon(\cdot)$ allows for the presence of unreliable source data, as it accounts for situations where η^P may not consistently provide a strong signal compared to η^Q . Additionally, $\varepsilon(\cdot)$ controls the ambiguity with respect to the distance from $1/2$. The larger the ambiguity level, the harder the classification task. Note that by the margin assumption, a trivial upper bound of the ambiguity level is $\varepsilon(z) = C_\alpha z^{1+\alpha}$.

2.3. On the ambiguity level. To help clarify the ambiguity level assumption, we provide some examples where explicit formulas for the ambiguity level $\varepsilon(\cdot)$ are available, with a proper choice of γ and C_γ . We defer the case of two logistic regression models across the target and source distributions to Section 5.

EXAMPLE 1 (Perfect source). Assume the condition of *relative signal exponent* proposed in Cai and Wei (2021), which amounts to

$$s(x) \geq C_\gamma \left| \eta^Q(x) - \frac{1}{2} \right|^\gamma \quad \forall x \in \Omega.$$

Then Assumption 2 holds with $\varepsilon(z) \equiv 0$.

Specifically, if $\eta^P = \eta^Q$, it is straightforward to set $\gamma = 1$ and $\varepsilon(\cdot) = 0$. In this scenario, the Bayes classifiers are identical and η^P gives strong signal compared to η^Q over the whole support Ω , so the ambiguity level is set as zero.

EXAMPLE 2 (Strong Signal over Ω_P). Instead of having a strong signal over the entire region as in Example 1, here we have a strong signal over only a given region:

$$s(x) \geq C_\gamma \left| \eta^Q(x) - \frac{1}{2} \right|^\gamma \quad \forall x \in \Omega_P.$$

Then Assumption 2 holds by taking

$$\varepsilon(z) \geq \mathbb{E}_{(X,Y) \sim Q} \left[\left| \eta^Q(X) - \frac{1}{2} \right| \mathbf{1} \left\{ \left| \eta^Q(X) - \frac{1}{2} \right| \leq z, X \notin \Omega_P \right\} \right],$$

that is, the ambiguity level is controlled by the risk within the complement Ω/Ω_P . This corresponds to the common case where the source data is collected from a subpopulation with respect to the target distribution.

EXAMPLE 3 (Strong Signal with Imperfect Transfer). Suppose that the transfer signal is strong, but the Bayes classifiers may differ, that is,

$$\left| \eta^P(x) - \frac{1}{2} \right| \geq C_\gamma \left| \eta^Q(x) - \frac{1}{2} \right|^\gamma, \quad \forall x \in \Omega,$$

$$\Omega_R := \left\{ x \in \Omega : \left(\eta^P(x) - \frac{1}{2} \right) \left(\eta^Q(x) - \frac{1}{2} \right) < 0 \right\} \neq \emptyset.$$

Then Assumption 2 holds if

$$\varepsilon(z) \geq \mathbb{E}_{(X,Y) \sim Q} \left[\left| \eta^Q(X) - \frac{1}{2} \right| \mathbf{1} \left\{ \left| \eta^Q(X) - \frac{1}{2} \right| \leq z, X \in \Omega_R \right\} \right].$$

In other words, the ambiguity level is determined by the risk associated with the region where different Bayes classifiers exist. A specific scenario that aligns with this example and utilizes $\gamma = C_\gamma = 1$ can be expressed as

$$\eta^P(x) = \eta^Q(x) \mathbf{1}\{x \notin \Omega_R\} + (1 - \eta^Q(x)) \mathbf{1}\{x \in \Omega_R\}, \quad \forall x \in \Omega,$$

where only the response values within Ω_R are flipped.

EXAMPLE 4 (Band-like Ambiguity). A further noteworthy scenario is when the probability distribution η^P concentrates around a “band” that is centered on an informative curve with respect to η^Q , but with some small deviation. A related situation is studied in [Reeve, Cannings and Samworth \(2021\)](#), where η^P is approximated by a linear transfer function of η^Q . Suppose that there exists some band error constant $\Delta \geq 0$, which represents the deviation level, such that

$$(4) \quad s(x) \geq C_\gamma \left| \eta^Q(x) - \frac{1}{2} \right|^\gamma - \Delta,$$

for any $x \in \Omega$. Then Assumption 2 holds with

$$(5) \quad \varepsilon(z; \gamma, C_\gamma/2) = (C_\alpha z^{1+\alpha}) \wedge (2^{\frac{1+\alpha}{\gamma}} C_\alpha C_\gamma^{-\frac{1+\alpha}{\gamma}} \Delta^{\frac{1+\alpha}{\gamma}}).$$

Notably, the case of $\Delta = 0$ reduces to the perfect source scenario in Example 1. For the proof of the statement in Example 4; see the Supplementary Material ([Fan, Gao and Klusowski \(2025\)](#)). For simplicity, we assume that (4) holds over the entire feature space Ω .

Specifically, if we further assume

$$(6) \quad \sup_{x \in \Omega} |\eta^P(x) - \eta^Q(x)| \leq \Delta,$$

then the band-like ambiguity condition (4) holds with $\gamma = C_\gamma = 1$. This condition is common and meaningful in reality because it reflects a scenario where, despite differences in data collection contexts, the underlying relationship between features and outcomes remains nearly identical across domains. By (5), Assumption 2 holds in this special case with

$$\varepsilon(z, 1, 1/2) = (C_\alpha z^{1+\alpha}) \wedge (2^{1+\alpha} C_\alpha C_\gamma^{-(1+\alpha)} \Delta^{1+\alpha}).$$

3. General convergence results. Let Π be any given subset of distributions (Q, P) satisfying that Assumptions 1 and 2 with parameter $\alpha \geq 0$ and $\gamma, C_\alpha, C_\gamma > 0$ and ambiguity level $\varepsilon(\cdot)$. Our analysis focuses on the performance of any classifier when the target and source distribution pair (Q, P) belongs to Π . This framework captures the essential information in the source data and how it can be used to improve the convergence rate of the excess risk.

3.1. Performance of the TAB model. In preparation for the general result, it is necessary to define the risk learned by the source data over the region $s(x) \geq C_\gamma |\eta^Q(x) - 1/2|^\gamma$ with strong signal strength.

DEFINITION 2 (Signal Transfer Risk). Define the *signal transfer risk* of the classifier f with respect to parameters $\gamma, C_\gamma > 0$ as

$$(7) \quad \xi(f; \gamma, C_\gamma) := \mathbb{E}_{(X,Y) \sim Q} \left[\left| \eta^Q(X) - \frac{1}{2} \right| \mathbf{1} \left\{ f(X) \neq f_Q^*(X), s(X) \geq C_\gamma \left| \eta^Q(X) - \frac{1}{2} \right|^\gamma \right\} \right].$$

We abbreviate it as $\xi(f)$ when there is no need to specify γ and C_γ .

The signal transfer risk results from the classification of points belonging to the area where $s(x) \geq C_\gamma |\eta^Q(x) - 1/2|^\gamma$. For simplicity, we assume the mild condition that $\sup_{(Q,P) \in \Pi} \mathbb{E}_{\mathcal{D}_P} \xi(\hat{f}^P) \gtrsim n_P^{-c}$ for some constant $c > 0$ to prevent a convergence rate faster than polynomial. Due to the strong signals offered by the source data within this area, it is intuitively expected that the signal transfer risk can be reduced with the aid of the source data sample $\mathcal{D}_P$.

In this paper, the classifier derived from the target data is assumed to be a *plug-in rule* of the form $\mathbf{1}\{\hat{\eta}^Q(x) \geq 1/2\}$, where $\hat{\eta}^Q$ is an estimator of the regression function η^Q . By introducing a novel strategy called the *TAB model*, the following result demonstrates a general approach to achieving a faster convergence rate of the excess risk.

THEOREM 1. *Let $\hat{\eta}^Q$ be an estimate of the regression function η^Q and $\hat{f}^P$ be a classifier obtained by $\mathcal{D}_P$. Suppose there exist two sequences δ_Q, δ_f such that $\delta_Q^{1+\alpha} \gtrsim n_Q^{-c}$ for some constant $c > 0$ and α defined by the margin Assumption 1 and that with probability at least $1 - \delta_f$, for any $x \in \Omega$ we have $\forall t > 0$,*

$$(8) \quad \sup_{(Q,P) \in \Pi} \mathbb{P}_{\mathcal{D}_Q}(|\hat{\eta}^Q(x) - \eta^Q(x)| \geq t | X_{1:n_Q}) \leq C_1 \exp\left(-\left(\frac{t}{\delta_Q}\right)^2\right),$$

for some constant $C_1 > 0$. Given the choice of $\tau \gtrsim \log(n_Q \vee n_P)\delta_Q$, the TAB classifier

$$(9) \quad \hat{f}_{\text{TAB}}(x) = \begin{cases} \mathbf{1}\left\{\hat{\eta}^Q(x) \geq \frac{1}{2}\right\}, & \text{if } \left|\hat{\eta}^Q(x) - \frac{1}{2}\right| \geq \tau, \\ \hat{f}^P(x), & \text{otherwise,} \end{cases}$$

satisfies, under the margin Assumption 1,

$$(10) \quad \sup_{(Q,P) \in \Pi} \mathbb{E}_{(\mathcal{D}_Q, \mathcal{D}_P)} \mathcal{E}_Q(\hat{f}_{\text{TAB}}) \lesssim \left(\sup_{(Q,P) \in \Pi} \mathbb{E}_{\mathcal{D}_P} \xi(\hat{f}^P) + \varepsilon(2\tau) \right) \wedge \tau^{1+\alpha} + \delta_f.$$

We now explain every important term in the upper bound (10).

- The term $\sup_{(Q,P) \in \Pi} \mathbb{E}_{\mathcal{D}_P} \xi(\hat{f}^P)$ captures the risk transferred by the source data using a classifier $\hat{f}^P$, excluding the ambiguity component. It often exhibits a faster convergence rate than $\delta_Q^{1+\alpha}$ and quantifies the benefits of transfer learning.
- The term $\varepsilon(2\tau)$ quantifies the ambiguity level *only when η^Q is close to 1/2*. Thus, the target data not only establishes an upper bound on the excess risk but also plays a critical role in reducing the risk caused by ambiguity in the source data.
- The threshold τ balances the classification ability of the target and source data by filtering out points easily classified with the target data. For $\tau = 0$, our approach mirrors [Audibert and Tsybakov \(2007\)](#), achieving an asymptotically lower excess risk than $\delta_Q^{1+\alpha}$ without source data. Alternatively, for $\tau = 1/2$, only the source data is used to construct the classifier, which is often the case in domain adaptation when no target data is available. Our choice of τ between these extremes combines the advantages of the target and source data.
- The final term δ_f represents the probability of the concentration inequality (8) failing due to extreme realizations of $\mathcal{D}_Q$. Generally, it is significantly smaller than the other terms that upper bound the excess risk. For instance, it decays exponentially with respect to n_Q in nonparametric classification using k -NN estimators (See Lemma 9.1 of [Cai and Wei \(2021\)](#)).

With the tight choice of $\tau \asymp \log(n_Q \vee n_P)\delta_Q$ to asymptotically minimize the right-hand side of (10), our upper bound cannot be worse than the rate $\delta_Q^{1+\alpha}$ obtained in [Audibert and Tsybakov \(2007\)](#), up to some logarithmic terms. This can happen when the transfer signal is too weak so that $\tau^{1+\alpha}$ dominates in (10).

Occasionally, the concentration of $\hat{\eta}^Q$ may not be exponential, as in (8). To overcome this limitation, the following theorem generalizes Theorem 1 to allow any type of concentration property of $\hat{\eta}^Q$.

THEOREM 2. *Let $\hat{\eta}^Q$ be an estimate of the regression function η^Q , and let $\hat{f}^P$ be a classifier obtained from $\mathcal{D}_P$. Suppose that for some $\tau > 0$, there exist functions $\delta(\cdot, \cdot)$, $\delta_Q(\cdot, \cdot)$ such that for any $(Q, P) \in \Pi$, the concentration property*

$$(11) \quad \mathbb{P}_{\mathcal{D}_Q}(|\hat{\eta}^Q(x) - \eta^Q(x)| \geq \tau) \leq \delta_Q(n_Q, \tau)$$

holds for any $x \in \Omega^ \subset \Omega$ with $Q(\Omega^*) \geq 1 - \delta(n_Q, \tau)$. Then the TAB classifier*

$$(12) \quad \hat{f}_{\text{TAB}}(x) = \begin{cases} \mathbf{1}\left\{\hat{\eta}^Q(x) \geq \frac{1}{2}\right\}, & \text{if } \left|\hat{\eta}^Q(x) - \frac{1}{2}\right| \geq \tau, \\ \hat{f}^P(x), & \text{otherwise,} \end{cases}$$

satisfies

$$(13) \quad \begin{aligned} & \sup_{(Q, P) \in \Pi} \mathbb{E}_{(\mathcal{D}_Q, \mathcal{D}_P)} \mathcal{E}_Q(\hat{f}_{\text{TAB}}) \\ & \lesssim \left(\sup_{(Q, P) \in \Pi} \mathbb{E}_{\mathcal{D}_P} \xi(\hat{f}^P) + \varepsilon(2\tau) \right) \wedge \tau^{1+\alpha} + \delta_Q(n_Q, \tau) + \delta(n_Q, \tau). \end{aligned}$$

This theorem holds for any given τ . The quantity $\delta_Q(n_Q, \tau)$ in (11) controls the failure probability bound of the concentration inequality concerning $\mathcal{D}_Q$. In Theorem 1, it is further incorporated into the probability of extreme realizations of $X_{1:n_Q}$ (8), and the subsequent concentration inequality for the failure probability related to $Y_{1:n_Q}$. To reduce the more general and abstract Theorem 2 to Theorem 1, it suffices to set $\delta_Q(n_Q, \tau) = \delta_f + C_1 \exp(-(\tau/\delta_Q)^2)$ and $\Omega^* \equiv \Omega$, where δ_f and C_1 are the notation used in Theorem 1.

Readers may be concerned that the exponential concentration (8) might not hold for all points over Ω . Indeed, this is particularly true when the support Ω is not compact, and the concentration inequalities only hold within a part of Ω with high probability. However, we can address this by adding a small failure probability with exponential concentration, that is, $1 - Q_X(\Omega^*) \leq \delta(n_Q, \tau)$, to the risk bound given in (13).

As the final part of this section, we list some recommendations of data-driven strategies for choosing the threshold parameter τ :

1. *Simple Rule of Thumb:* We recommend any value between $0.5n_Q^{-1/2}$ and $n_Q^{-1/2}$.
2. *Adaptive Selection of Base Model Parameters:* If δ_Q is represented by parameters in specific base models, adjust τ based on these parameters. For example, Sections 4 and 5 show that δ_Q is directly related to the choice of nearest neighbors and the lasso penalty parameter.
3. *Empirical Risk Minimization:* Select τ by minimizing the empirical risk over the target data, choosing the one with the best empirical loss from a candidate list of τ .

A detailed discussion of these approaches, including a rigorous analysis of the theoretical behavior of ERM and the computationally feasible ERM algorithm, is provided in the Supplementary Material (Fan, Gao and Klusowski (2025)).

3.2. Simple approach to bounding signal transfer risk. The majority of the existing literature focuses on bounding the target excess risk using only the target data. While the excess risk of a classifier $\hat{f}^P$ with respect to the source distribution, that is,

$$2\mathbb{E}_{(X, Y) \sim P} \left[\left| \eta^P(X) - \frac{1}{2} \right| \mathbf{1}\{\hat{f}^P(X) \neq f_P^*(X)\} \right],$$

is well studied, its signal transfer risk $\xi(\hat{f}^P)$, given by (7), is less explored.

To incorporate the vast literature of traditional statistical learning into our framework, we present a result that directly bounds the signal transfer risk $\xi(\hat{f}^P)$ in terms of the excess risk

of the source distribution. This result involves a more refined version of the signal transfer risk, and more significantly, does not rely on any concentration property often required by plug-in rules.

Some additional assumptions are needed. Condition 1 ensures the boundedness of the Radon–Nikodym derivative $\frac{dQ_X}{dP_X}$ over $\Omega \cap \Omega_P$. This ensures that the source distribution provides enough coverage over $\Omega \cap \Omega_P$ to enable accurate learning of every point over the common supporting region relative to Q_X .

CONDITION 1 (Absolute Continuity). Q_X is absolutely continuous with respect to P_X . Moreover, there exists some constant $M > 0$ such that the Radon–Nikodym derivative satisfies $\frac{dQ_X}{dP_X}(x) \leq M$ for any $x \in \Omega \cap \Omega_P$.

Denote all such marginal distribution pairs (Q_X, P_X) that satisfy Condition 1 with parameter $M > 0$ by $\mathcal{A}(M)$. We are in a position to present the upper bound for the signal transfer risk with respect to the target distribution in terms of the source excess risk.

THEOREM 3. *Define the source excess risk as*

$$\varepsilon_P := \sup_{(Q, P) \in \Pi \cap \mathcal{A}(M)} \mathbb{E}_{\mathcal{D}_P} \mathbb{E}_{(X, Y) \sim P} \left[\left| \eta^P(X) - \frac{1}{2} \mathbf{1}\{\hat{f}^P(X) \neq f_P^*(X)\} \right| \right].$$

Then the signal transfer risk satisfies

$$\sup_{(Q, P) \in \Pi \cap \mathcal{A}(M)} \mathbb{E}_{\mathcal{D}_P} \xi(\hat{f}^P) \leq \begin{cases} 2M^{\frac{1+\alpha}{\gamma+\alpha}} C_\alpha^{\frac{\gamma-1}{\gamma+\alpha}} C_\gamma^{-\frac{1+\alpha}{\gamma+\alpha}} \varepsilon_P^{\frac{1+\alpha}{\gamma+\alpha}}, & \gamma \geq 1, \\ 2^{\gamma-1} M C_\gamma^{-1} \varepsilon_P, & \gamma < 1. \end{cases}$$

Although the result may be suboptimal when $\gamma < 1$ and additionally requires that $(Q, P) \in \mathcal{A}(M)$, Theorem 3 shows that the signal transfer risk can be bounded by the source excess risk ε_P , and inherits the performance and possible consistency with respect to classification on the source data. As long as $\hat{f}^P$ learns the source distribution sufficiently well, the signal transfer risk will converge faster than the rate obtained with only the target data, namely $\delta_Q^{1+\alpha}$.

Theorem 3 only requires knowledge of conventional statistical learning techniques, which are widely studied and well understood in the literature. Therefore, researchers and practitioners can easily apply our framework to a wide range of problems, without the need for specialized knowledge or expertise in transfer learning or related fields. Conventional theory on the excess risk can thus be applied to transfer learning.

4. Applications in nonparametric classification. In this section, we aim to apply the general result of Theorem 1 to nonparametric classification settings. Here, we design the TAB classifier by combining plug-in rules over the target and source data, and obtain min-max optimal rates under nonparametric settings. See Audibert and Tsybakov (2007) for a comprehensive overview of theoretical properties of plug-in rules.

We adopt K -nearest neighbor (K -NN) classifiers as plug-in rules for both $\hat{\eta}^Q$ and $\hat{\eta}^P$. Our analysis in this section then builds on prior work on rates for K -nearest neighbor classification (e.g., Hall, Park and Samworth (2008), Samworth (2012), Gadat, Klein and Marteau (2016), Celisse and Mary-Huard (2018), Cannings, Berrett and Samworth (2020)). For a review of early work on the theoretical properties of the K -NN classifier, see Devroye, Györfi and Lugosi (1996). Also, in the literature of nonparametric classification, see Fan (1993) and Fan and Gijbels (1996) for local polynomial regression as an alternative to K -NN methods.

If the classifier $\hat{f}^P(\cdot) = \mathbf{1}\{\hat{\eta}^P(\cdot) \geq 1/2\}$ is a general plug-in rule, we also provide an explicit upper bound for the signal transfer risk $\xi(\hat{f}^P)$ based on the pointwise misclassification rate of $\hat{\eta}^P(x)$. See the Supplementary Material (Fan, Gao and Klusowski (2025)) for the details of this bound and related results.

4.1. *K-nearest neighbor TAB classifier.* Given a query point $x \in \mathbb{R}^d$, we first reorder the target data pairs as $(X_{(1)}, Y_{(1)}), \dots, (X_{(n_Q)}, Y_{(n_Q)})$ based on the Euclidean distances of the X_i 's to x , that is,

$$\|X_{(1)} - x\|_2 \leq \dots \leq \|X_{(n_Q)} - x\|_2.$$

Then we define the K -NN estimate $\hat{\eta}_{k_Q}^Q(x)$ as the simple average of the response values of the k_Q nearest neighbors of x in the target data:

$$\hat{\eta}_{k_Q}^Q = \frac{1}{k_Q} \sum_{i=1}^{k_Q} Y_{(i)}(x).$$

Similarly, we define the K -NN estimate $\hat{\eta}_{k_P}^P(x) := k_P^{-1} \sum_{i=1}^{k_P} Y_{(i)}^P(x)$ for the source data pairs $\mathcal{D}_P$. Finally, we plug these estimates into the TAB K -NN classifier to obtain

$$\hat{f}_{\text{TAB}}^{\text{NN}}(x) = \begin{cases} \mathbf{1} \left\{ \hat{\eta}_{k_Q}^Q(x) \geq \frac{1}{2} \right\}, & \text{if } \left| \hat{\eta}_{k_Q}^Q(x) - \frac{1}{2} \right| \geq \tau, \\ \mathbf{1} \left\{ \hat{\eta}_{k_P}^P(x) \geq \frac{1}{2} \right\}, & \text{otherwise.} \end{cases}$$

As for the threshold parameter, we choose

$$\tau \asymp \log(n_Q \vee n_P) k_Q^{-\frac{1}{2}}.$$

Further elaboration on the rationale behind choosing τ , as well as the well-studied optimal selection of (k_Q, k_P) , can be found in Section 4.3.

4.2. *Nonparametric classification setting.* We are now in a position to state the applications of our proposed TAB classifier in nonparametric classification under the finite dimension regime. In addition to the margin assumption and the ambiguity level assumption, this paper considers the nonparametric classification problem when the following smoothness condition holds.

CONDITION 2 (Smoothness). For any $\beta \in [0, 1]$, the (β, C_β) -Hölder is the class of functions $g : \mathbb{R}^d \rightarrow \mathbb{R}$ satisfying, for any $x, x' \in \mathbb{R}^d$,

$$|g(x) - g(x')| \leq C_\beta \|x - x'\|^\beta,$$

for some constant $C_\beta > 0$. We denote this class of functions by $\mathcal{H}(\beta, C_\beta)$.

Previous works, including Cai and Wei (2021) and Reeve, Cannings and Samworth (2021), do not typically require any smoothness assumption for η^P . While this simplification is justified when η^P is closely related to η^Q , it may overlook valuable information from a sufficiently smooth η^P , leading to a phase-transition in the upper and lower bounds of the excess risk (for a detailed discussion, refer to the Supplementary Material (Fan, Gao and Klusowski (2025))). In contrast, our approach considers the smoothness of both the target and source regression functions. Specifically, we assume that

$$\eta^Q \in \mathcal{H}(\beta, C_\beta), \quad \eta^P \in \mathcal{H}(\beta_P, C_{\beta_P}).$$

This condition allows us to obtain a more refined upper bound that depends on the smoothness of both functions.

Our next condition concerns the mass of the source and target distributions in the sense that the density functions with respect to Q_X and P_X are bounded from zero and infinity. A similar condition has been imposed in (Audibert and Tsybakov (2007)) and Cai and Wei (2021). We require that both Q_X and P_X satisfy the following condition.

CONDITION 3 (Strong Density). The marginal distribution Q_X is absolutely continuous with respect to the Lebesgue measure λ on its *compact support* (denoted by Ω). Furthermore, we have that

$$\frac{dQ_X}{d\lambda}(x) \in [\mu^-, \mu^+], \quad \frac{\lambda(B(x, r) \cap \Omega)}{\lambda(B(x, r))} \geq c_\mu, \quad \forall 0 < r < r_u, x \in \Omega,$$

Denote the set of such marginal distributions Q_X by $\mathcal{S}(\mu)$ with positive parameters $\mu = (\mu^+, \mu^-, c_\mu, r_\mu)$.

Taking all the conditions above into account, we consider the subset of source and target distribution pairs in Π that satisfies Assumptions 1, 2 and Conditions 2, 3:

$$\Pi^{\text{NP}} := \{(Q, P) \in \Pi : \eta^Q \in \mathcal{H}(\beta, C_\beta), \eta^P \in \mathcal{H}(\beta_P, C_{\beta_P}), Q_X, P_X \in \mathcal{S}(\mu)\}.$$

We further impose the mild assumption $\alpha\beta \leq d$ to rule out the “superfast” rates of convergence mentioned in Audibert and Tsybakov (2007). This is guaranteed to hold when η^Q equals $1/2$ at an interior point of Ω (See Proposition 3.4 of Audibert and Tsybakov (2007)).

In the following part of this section, we explore three types of additional conditions on the space Π^{NP} and analyze the respective excess risks.

1. *Band-like Ambiguity*: We consider the scenario of band-like ambiguity described in Example 4. Define the focal parametric space as

$$\Pi_{\text{BA}}^{\text{NP}} := \left\{ (Q, P) \in \Pi^{\text{NP}} : s(x) \geq C_\gamma \left| \eta^Q(x) - \frac{1}{2} \right|^\gamma - \Delta, \quad \forall x \in \Omega \right\}.$$

Recall that in Example 4, we had shown that this implies

$$\varepsilon(z; \gamma, C_\gamma/2) = (C_\alpha z^{1+\alpha}) \wedge (2^{\frac{1+\alpha}{\gamma}} C_\alpha C_\gamma^{-\frac{1+\alpha}{\gamma}} \Delta^{\frac{1+\alpha}{\gamma}}).$$

Furthermore, in this case, we set $\beta_P = 0$, which means that no additional smoothness condition is imposed on η^P . Instead, we allow η^P to arbitrarily fluctuate within a small band whose width is measured by Δ . When $\Delta = 0$, our setting reduces to the one in Cai and Wei (2021).

2. *Smooth Source*: We add the condition $\beta_P \geq \gamma\beta$ for scenarios where η^P is smooth. While the ambiguity level $\varepsilon(\cdot)$ is arbitrary, this condition ensures that points with strong signal strength have neighboring data points with strong signal strength of η^P as well, enhancing the classification of source data points with strong signal strength.

Define Π_S^{NP} as the subset of Π^{NP} such that $\beta_P \geq \gamma\beta$. When η^Q and η^P share the same smoothness degree β , we set $\gamma = 1$.

3. *Strong Signal with Imperfect Transfer*: We consider the scenario in Example 3 where a strong signal strength exists but the direction may be reversed. This condition ensures that the region $s(x) \leq C_\gamma |\eta^Q(x) - 1/2|^\gamma$ is smooth, which further ensures the availability of a sufficient number of neighboring source data points with the strong signal. In fact, its boundary is a part of the decision boundary $\{x \in \Omega : \eta^Q(x) = 1/2\}$ whose smoothness is guaranteed by the smoothness of η^Q (see the proof of Theorem 4).

Define

$$\Pi_I^{\text{NP}} := \left\{ (Q, P) \in \Pi^{\text{NP}} : \eta^P \text{ is continuous, } \left| \eta^P(x) - \frac{1}{2} \right| \geq C_\gamma \left| \eta^Q(x) - \frac{1}{2} \right|^\gamma, \quad \forall x \in \Omega \right\}.$$

It is worth noting that $s(x) \geq C_\gamma |\eta^Q(x) - 1/2|^\gamma$ if and only if $x \notin \Omega_R$, that is, $(\eta^P(x) - 1/2)(\eta^Q(x) - 1/2) \geq 0$. Therefore, the ambiguity level

$$\varepsilon(z) = \mathbb{E}_{(X,Y) \sim Q} \left[\left| \eta^Q(X) - \frac{1}{2} \right| \mathbf{1} \left\{ \left| \eta^Q(X) - \frac{1}{2} \right| \leq z, X \in \Omega_R \right\} \right]$$

precisely captures the risk caused by different Bayes classifiers between the target and source data. We also let $\beta_P = 0$ in this case, but assume that η^P is continuous to make sure that the region $s(x) \leq C_\gamma |\eta^Q(x) - 1/2|^\gamma$ has a continuous boundary.

The analysis of the excess risk rate with respect to the wider class Π^{NP} is provided in the Supplementary Material (Fan, Gao and Klusowski (2025)).

4.3. Optimal rate of excess risk. Before presenting the results regarding the optimal rate of $\mathcal{E}_Q(\hat{f}_{\text{TAB}}^{\text{NN}})$, we first discuss parameter selection. We choose the number of nearest neighbors as follows:

$$k_Q = \lfloor c_Q n_Q^{\frac{2\beta}{2\beta+d}} \rfloor, \quad k_P = \lfloor c_P n_P^{\frac{2\gamma\beta}{2\gamma\beta+d}} \rfloor,$$

where c_Q and c_P are positive constants. This choice is motivated by previous work such as Gadat, Klein and Marteau (2016) and Cannings, Berrett and Samworth (2020), where similar choices are made in the context of nearest-neighbor methods. Notably, the choice of k_P is similarly derived by seeing $\gamma\beta$ as the “smoothness” parameter for the source data, and our choice coincides with the classical optimal choice when $\beta_P = \gamma\beta$. In addition, we assume γ and β are known here for convenience. For adaptive and rate-optimal approaches to determining the number of nearest neighbors, see Lepskiĭ (1992), Reeve, Cannings and Samworth (2021).

As for the threshold in the TAB classifier $\hat{f}_{\text{TAB}}^{\text{NN}}$, we choose

$$\tau \asymp \log(n_Q \vee n_P) k_Q^{-\frac{1}{2}}.$$

This choice is consistent with the concentration property of $\hat{\eta}_{k_Q}^Q$, of which the “uncertainty” level δ_Q in (8) is proportional to $k_Q^{-\frac{1}{2}}$, since $\hat{\eta}_{k_Q}^Q$ is the average of k_Q random variables. By the definition of k_Q , we see that

$$\tau \asymp \log(n_Q \vee n_P) n_Q^{-\frac{\beta}{2\beta+d}}.$$

Given a family of the target and source distributions, the next theorem gives a provable upper bound on the excess risk of the TAB K -NN classifier $\hat{f}_{\text{TAB}}^{\text{NN}}$, with the proper parameter choices.

THEOREM 4 (Nonparametric Classification Upper Bound). *Suppose that $n_Q^{\frac{d}{2\beta+d}} \times \exp(-c_Q n_Q^{\frac{2\beta}{2\beta+d}}) \lesssim n_P^{-\frac{\beta(1+\alpha)}{2\gamma\beta+d}}$. Then the TAB K -NN classifier $\hat{f}_{\text{TAB}}^{\text{NN}}(x)$ satisfies:*

1. Band-like Ambiguity:

$$(14) \quad \sup_{(Q,P) \in \Pi_{\text{BA}}^{\text{NP}}} \mathbb{E}_{(\mathcal{D}_Q, \mathcal{D}_P)} \mathcal{E}_Q(\hat{f}_{\text{TAB}}^{\text{NN}}) \lesssim (n_P^{-\frac{\beta(1+\alpha)}{2\gamma\beta+d}} + \Delta^{\frac{1+\alpha}{\gamma}}) \wedge (\log^{1+\alpha}(n_Q \vee n_P) n_Q^{-\frac{\beta(1+\alpha)}{2\beta+d}}).$$

2. Smooth Source:

$$(15) \quad \sup_{(Q,P) \in \Pi_S^{\text{NP}}} \mathbb{E}_{(\mathcal{D}_Q, \mathcal{D}_P)} \mathcal{E}_Q(\hat{f}_{\text{TAB}}^{\text{NN}}) \lesssim (n_P^{-\frac{\beta(1+\alpha)}{2\gamma\beta+d}} + \varepsilon(2\tau)) \wedge (\log^{1+\alpha}(n_Q \vee n_P) n_Q^{-\frac{\beta(1+\alpha)}{2\beta+d}}).$$

3. Strong Signal with Imperfect Transfer:

$$(16) \quad \sup_{(Q,P) \in \Pi_I^{\text{NP}}} \mathbb{E}_{(\mathcal{D}_Q, \mathcal{D}_P)} \mathcal{E}_Q(\hat{f}_{\text{TAB}}^{\text{NN}}) \lesssim (n_P^{-\frac{\beta(1+\alpha)}{2\gamma\beta+d}} + \varepsilon(2\tau)) \wedge (\log^{1+\alpha}(n_Q \vee n_P) n_Q^{-\frac{\beta(1+\alpha)}{2\beta+d}}).$$

Theorem 4 is obtained simply by verifying the conditions in Theorem 1, as detailed in the Supplementary Material (Fan, Gao and Klusowski (2025)). The Supplementary Material presents a more refined and technical approach to analyze the signal transfer risk of plug-in rules compared to Theorem 3. The mild condition $n_Q^{\frac{d}{2\beta+d}} \exp(-c_Q n_Q^{\frac{2\beta}{2\beta+d}}) \lesssim n_P^{-\frac{\beta(1+\alpha)}{2\gamma\beta+d}}$ controls the probability of failure δ_f in the terminology of Theorem 1.

Our risk bounds reveal that transfer learning leads to faster convergence rates of excess risk when n_P is large compared to n_Q and the ambiguity level is small; specifically,

$$\begin{aligned} \Pi_{BA}^{NP} : n_P \gg n_Q^{\frac{2\gamma\beta+d}{2\beta+d}}, \Delta \ll n_Q^{-\frac{\gamma\beta}{2\beta+d}}; \\ \Pi_S^{NP}, \Pi_I^{NP} : n_P \gg n_Q^{\frac{2\gamma\beta+d}{2\beta+d}}, \varepsilon(2\tau) \ll \tau^{1+\alpha}. \end{aligned}$$

On the other hand, if n_P is small compared to n_Q , then the term $n_Q^{-\frac{\beta(1+\alpha)}{2\beta+d}}$ dominates the upper bound, and reduces to the risk rate in the conventional setting with only target data and the strong density assumption (Audibert and Tsybakov (2007)), up to logarithmic factors.

REMARK 1. (a) When $\Delta = 0$ in (14), or $\varepsilon(\cdot) \equiv 0$ in (16), our upper bound reduces to Theorem 2 of Cai and Wei (2021), that is, $n_P^{-\frac{\beta(1+\alpha)}{2\gamma\beta+d}} \wedge n_Q^{-\frac{\beta(1+\alpha)}{2\beta+d}}$, up to logarithmic factors. Importantly, our method demonstrates robustness against a positive band error Δ , in contrast to the weighted K -NN estimator proposed in their work. In addition, our proposed TAB method is simple and does not require the careful design of the classifier, such as the weighting or Lepski's method used in Cai and Wei (2021) and Reeve, Cannings and Samworth (2021). We refer the reader to Figure 2 in Section 6 for a numerical comparison.

(b) Our result (15) supplements these previous works (Cai and Wei (2021), Reeve, Cannings and Samworth (2021)) by allowing for an arbitrary type of ambiguity after imposing a smoothness condition on η^P .

We next provide lower bounds on the minimax excess risk, which show that the TAB K -NN classifier achieves the minimax optimal rates, even when we add the constraint $\Omega = \Omega_P$. All lower bound results are constructed over a subset of Π^{NP} , which is also a subset of the distribution pair family satisfying Assumptions 1 and 2 with specified ambiguity level $\varepsilon(\cdot)$.

THEOREM 5 (Nonparametric Classification Lower Bound). *Fix the parameters $\alpha\beta \leq d$ and set $\tau \asymp \log(n_Q \vee n_P) n_Q^{-\frac{\beta}{2\beta+d}}$. We have that*

1. Band-like Ambiguity:

$$(17) \quad \inf_{\hat{f}} \sup_{\substack{(Q,P) \in \Pi_{BA}^{NP} \\ \Omega = \Omega_P}} \mathbb{E}_{(\mathcal{D}_Q, \mathcal{D}_P)} \mathcal{E}_Q(\hat{f}) \gtrsim (n_P^{-\frac{\beta(1+\alpha)}{2\gamma\beta+d}} + \Delta^{\frac{1+\alpha}{\gamma}}) \wedge n_Q^{-\frac{\beta(1+\alpha)}{2\beta+d}}.$$

2. Smooth Source: For some constant $c > 0$ that is independent of α ,

$$(18) \quad \inf_{\hat{f}} \sup_{\substack{(Q,P) \in \Pi_S^{NP} \\ \Omega = \Omega_P}} \mathbb{E}_{(\mathcal{D}_Q, \mathcal{D}_P)} \mathcal{E}_Q(\hat{f}) \gtrsim (n_P^{-\frac{\beta(1+\alpha)}{2\gamma\beta+d}} + \varepsilon(cn_Q^{-\frac{\beta}{2\beta+d}})) \wedge n_Q^{-\frac{\beta(1+\alpha)}{2\beta+d}}.$$

3. Strong Signal with Imperfect Transfer: For some constant $c > 0$ that is independent of α ,

$$(19) \quad \inf_{\hat{f}} \sup_{\substack{(Q,P) \in \Pi_I^{NP} \\ \Omega = \Omega_P}} \mathbb{E}_{(\mathcal{D}_Q, \mathcal{D}_P)} \mathcal{E}_Q(\hat{f}) \gtrsim (n_P^{-\frac{\beta(1+\alpha)}{2\gamma\beta+d}} + \varepsilon(cn_Q^{-\frac{\beta}{2\beta+d}})) \wedge n_Q^{-\frac{\beta(1+\alpha)}{2\beta+d}}.$$

In the special case where $\sup_{x \in \Omega} |\eta^Q(x) - \eta^P(x)| \leq \Delta$, we can determine that the minimax optimal excess risk is $(n_P^{-\frac{\beta(1+\alpha)}{2\beta+d}} + \Delta^{1+\alpha}) \wedge n_Q^{-\frac{\beta(1+\alpha)}{2\beta+d}}$, up to logarithmic factors of $n_Q \vee n_P$. As long as

$$n_P \gg n_Q, \quad \Delta \ll n_Q^{-\frac{\beta}{2\beta+d}},$$

the classifier will benefit from the source data. The proof of Theorem 5 reveals that condition (6), although slightly stronger than the band-like ambiguity condition (4), remains compatible with the lower bound construction as it ensures $\sup_{x \in \Omega} |\eta^Q(x) - \eta^P(x)| \leq \Delta$.

5. Applications in logistic regression. Besides nonparametric classification, we also investigate the use of transfer learning in logistic regression models, which are a commonly used parametric approach in classification. Previous works such as Zheng et al. (2019) have studied the “data enriched model” for logistic regression under a single-source setting, Abramovich and Grinshtein (2019) have explored sparse logistic regression in high-dimensional settings, and Tian and Feng (2023) have considered transfer learning in generalized linear models. Our goal is to reveal how incorporating an additional source logistic regression model with a different linear regression coefficient can enhance the convergence of the excess misclassification rate.

In this section, we extend to the high-dimensional regime, allowing $d \rightarrow \infty$ as $n_Q \rightarrow \infty$. Suppose the source and target distributions are logistic regression models given by

$$(20) \quad \begin{aligned} \text{Target data model: } \eta^Q(x) &= \sigma(\beta_Q^T x), \\ \text{Source data model: } \eta^P(x) &= \sigma(\beta_P^T x), \end{aligned}$$

where two independent samples $(X_1, Y_1), \dots, (X_{n_Q}, Y_{n_Q}) \stackrel{\text{i.i.d.}}{\sim} Q$ and $(X_1^P, Y_1^P), \dots, (X_{n_P}^P, Y_{n_P}^P) \stackrel{\text{i.i.d.}}{\sim} P$ are observed. To simplify the theoretical analysis, we assume that the marginal distributions Q_X and P_X are both $N(0, I_d)$, the d -dimension standard normal distribution. This marginal distribution is convenient when working with the restricted strong convexity condition (See Negahban et al. (2009)). It can be extended to the more general distributions studied by Dobriban and Fan (2016).

Let $\angle(\alpha, \beta)$ be the angle of two vectors α and β , in the range of 0 and $\pi/2$. Consider the following parametric space of the coefficient pair (β_Q, β_P) :

$$(21) \quad \Theta(s, \Delta) = \{(\beta_Q, \beta_P) : \|\beta_Q\|_0 \leq s, \angle(\beta_Q, \beta_P) \leq \Delta\},$$

for some $s > 0$ and $\Delta \in [0, \pi/2]$. The corresponding family of distribution pairs is then

$$(22) \quad \begin{aligned} \Pi^{\text{LR}} = \Pi^{\text{LR}}(s, \Delta, M) &= \{(Q, P) : X, X^P \sim N(0, I_d), \eta^Q(x) = \sigma(\beta_Q^T x), \\ &\eta^P(x) = \sigma(\beta_P^T x), (\beta_Q, \beta_P) \in \Theta(s, \Delta)\}. \end{aligned}$$

To ensure the control of the ambiguity level, we impose a constraint on the angle between β_Q and β_P in (21), which must be smaller than a constant Δ . Importantly, our constructed parametric space does not impose any sparsity conditions on β_P .

Given the family of logistic distribution pairs Π^{LR} , we show that Π^{LR} is a subset of the overall distribution pair space Π with $\alpha = 1$ and $\varepsilon(z, 1, m/\pi) \lesssim z^2 \wedge \Delta^2$, provided that $\|\beta_P\| \geq m\|\beta_Q\|$ for some constant $m > 0$. See the Supplementary Material (Fan, Gao and Klusowski (2025)) for a detailed proof.

For model fitting, we minimize the negative Bernoulli likelihood function with lasso regularization terms to obtain $\hat{\beta}_Q$ and $\hat{\beta}_P$, that is,

$$(23) \quad \begin{aligned} \hat{\beta}_Q &= \arg \min_{\beta \in \mathbb{R}^p} \frac{1}{n_Q} \sum_{i=1}^{n_Q} \{\log(1 + e^{X_i^T \beta}) - Y_i X_i^T \beta\} + \lambda_Q \|\beta\|_1, \\ \hat{\beta}_P &= \arg \min_{\beta \in \mathbb{R}^p} \frac{1}{n_P} \sum_{i=1}^{n_P} \{\log(1 + e^{X_i^T \beta}) - Y_i^P X_i^{P^T} \beta\} + \lambda_P \|\beta\|_1, \end{aligned}$$

where $\lambda_* \asymp \sqrt{\frac{\log d}{n_*}}$ for $* \in \{Q, P\}$. The corresponding target and source plug-in classifiers are then $\mathbf{1}\{\sigma(\hat{\beta}_Q^T x) \geq 1/2\} = \mathbf{1}\{\hat{\beta}_Q^T x \geq 0\}$ and $\mathbf{1}\{\hat{\beta}_P^T x \geq 0\}$. Hence, the TAB logistic lasso classifier becomes

$$\hat{f}_{\text{TAB}}^{\text{LR}}(x) = \begin{cases} \mathbf{1}\{\hat{\beta}_Q^T x \geq 0\}, & \text{if } |\sigma(\hat{\beta}_Q^T x) - 1/2| \geq \tau, \\ \mathbf{1}\{\hat{\beta}_P^T x \geq 0\}, & \text{otherwise.} \end{cases}$$

By setting λ_Q, λ_P , and τ properly, the following theorem gives an excess risk upper bound.

THEOREM 6. *Assume that $L \leq m \|\beta_Q\|_2 \leq \|\beta_P\|_2 \leq U$ for some constants $L, U > 0$ and $0 < m \leq 1$. Suppose that for some constant $K > 0$, we have $d^K \gtrsim \frac{n_Q \vee n_P}{s \log d}$, $n_Q \gg \log \frac{n_P}{s \log d}$, and $n_Q \wedge n_P \gg s \log d$. Let $\hat{\beta}_Q, \hat{\beta}_P$ be obtained in (23) with*

$$\lambda_Q = c_Q \sqrt{\frac{\log d}{n_Q}}, \quad \lambda_P = c_P \sqrt{\frac{\log d}{n_P}},$$

for some constants $c_Q, c_P \geq \sqrt{(K+1)}$. The TAB lasso classifier with threshold

$$\tau = c_\tau \sqrt{\frac{s \log d}{n_Q}} \log(n_Q \vee n_P),$$

for some constant $c_\tau > 0$, satisfies

$$(24) \quad \sup_{(Q, P) \in \Pi^{\text{LR}}} \mathbb{E}_{(\mathcal{D}_Q, \mathcal{D}_P)} \mathcal{E}_Q(\hat{f}_{\text{TAB}}^{\text{LR}}) \lesssim \left(\frac{s \log d}{n_P} + \Delta^2 \right) \wedge \left(\frac{s \log d}{n_Q} \log^2(n_Q \vee n_P) \right).$$

The term $\frac{s \log d}{n_Q}$ is the classical risk term with access to only the target data, and $\frac{s \log d}{n_P}$ is the risk term transferred by the source data with an additional term Δ^2 measuring the angle discrepancy between β_Q and β_P . We see that knowledge from the source data can significantly improve the learning performance when n_P is large and Δ is small, namely

$$n_P \gg n_Q, \quad \Delta \ll \sqrt{\frac{s \log d}{n_Q}}.$$

It is worth noting that when $\|\beta_Q\|$ and $\|\beta_P\|$ are bounded away from zero, the small angle condition between β_Q and β_P considered in this paper is a more general assumption than the contrast assumption in [Tian and Feng \(2023\)](#), which requires that β_Q is sparse and the l_q -norm of the difference between β_Q and β_P is small for some $q \in [0, 1]$. In contrast, our result is applicable to a broader class of parameter spaces, allowing for β_P to not only be nonsparse, but also differ significantly in norm from β_Q .

We choose this τ to ensure that, with high probability, we have $\|\hat{\beta}_Q - \beta_Q\|_2 \lesssim \sqrt{s} \lambda_Q$. This, in turn, implies

$$|\sigma(\hat{\beta}_Q^T x) - \sigma(\beta_Q^T x)| \lesssim \sqrt{s} \lambda_Q$$

with high probability with respect to $Q_X = N(0, I_d)$. The proof of Theorem 6 then comes from verifying that $\sup_{(Q,P) \in \Pi^{\text{LR}}} \mathbb{E}_{\mathcal{D}_P} \xi(\mathbf{1}\{\hat{\beta}_P^T x \geq 0\}) \asymp \frac{s \log d}{n_P} + \Delta^2$ and $\varepsilon(z) \asymp z^2 \wedge \Delta^2$ in Theorem 2. This indicates that even when β_P is nonsparse, we can still obtain a reliable estimate of β_P by incorporating a lasso regularizer, if in addition Δ is sufficiently small.

Theorem 7 below shows that the upper bound (24) in Theorem 6 is optimal up to logarithmic factors of $n_Q \vee n_P$.

THEOREM 7. *Suppose that $\frac{s \log d}{n_Q \vee n_P} \lesssim 1$. We have that*

$$\inf_{\hat{f}} \sup_{(Q,P) \in \Pi^{\text{LR}}} \mathbb{E} \mathcal{E}_Q(\hat{f}) \gtrsim \left(\frac{s \log d}{n_P} + \Delta^2 \right) \wedge \frac{s \log d}{n_Q}.$$

The derivation of the lower bound involves two terms. The term $\frac{s \log d}{n_P} \wedge \frac{s \log d}{n_Q}$ represents the optimal convergence rate when the target and source distributions are identical, that is, $\beta_P = \beta_Q$. The term $\Delta^2 \wedge \frac{s \log d}{n_Q}$ corresponds to the choice $\beta_P = (1, 0, 0, \dots, 0)$ in the lower bound construction, which imposes a sparsity constraint on β_Q within a small cone.

It is worth noting that traditional lower bounds, typically derived from Fano's lemma, only consider minimax rates with respect to a distance metric. To show our lower bound of the excess risk $\mathbb{E} \mathcal{E}_Q(\hat{f})$, we introduce a novel transformation that relates the excess risk to the angle difference of the linear coefficients. By applying Fano's lemma to this transformed quantity, we obtain the desired lower bound. We refer the reader to the Supplementary Material (Fan, Gao and Klusowski (2025)) for a detailed explanation.

6. Simulation studies. As mentioned earlier, the TAB classifier offers benefits in scenarios where n_P is large and the ambiguity level is small, and prevents negative transfer when the source data lacks sufficient information to aid the classification. In this section, we present simulation studies to demonstrate the practical benefits of transfer learning and the TAB classifier. We separately consider the nonparametric classification and logistic regression settings.

6.1. Nonparametric classification setting. The setting we consider is as follows: $d = 2$, $\Omega = \Omega_P = [0, 1]^2$, $Q_X = P_X = \text{Uniform}([0, 1]^2)$, the uniform distribution over the square. For the target regression function, let $\eta^Q(x) = \eta^Q(x_1, x_2) = \frac{1}{2} + \frac{1}{10} \sin(2\pi(x_1 + x_2))$. We have $\alpha = \beta = 1$ for some constants C_α and C_β . Next, we consider two different nonparametric regression scenarios, by specifying the source regression function η^P with different types of ambiguity.

1. *Band-like Ambiguity:* For $\Delta \in \{0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6\}$, $\gamma \in \{0.5, 1\}$:

$$\eta^P(x) = \frac{1}{2} + \left\{ 2 \left(\eta^Q(x) - \frac{1}{2} \right)^\gamma - \Delta \right\} \text{sgn} \left(\eta^Q(x) \geq \frac{1}{2} \right).$$

Here, η^P concentrates around an informative curve with respect to η^Q with some ambiguity Δ . In this case, we have $s(x) \geq |\eta^Q(x) - 1/2|^\gamma - \Delta$.

2. *Partially Flipped Sine Functions:* For $r \in \{0, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35\}$, $\gamma \in \{0.5, 1\}$:

$$\eta^P(x) = \eta^P(x_1, x_2) = \begin{cases} \frac{1}{2} - \frac{1}{5} \sin \left(2\pi \frac{\{x_1 + x_2\}}{r} \right)^\gamma, & \text{if } \{2x_1 + 2x_2\} \in [0, r], \\ \frac{1}{2} + \frac{1}{5} \sin \left(2\pi \frac{\{x_1 + x_2\} - r}{1 - r} \right)^\gamma, & \text{if } \{2x_1 + 2x_2\} \in (r, 1], \end{cases}$$

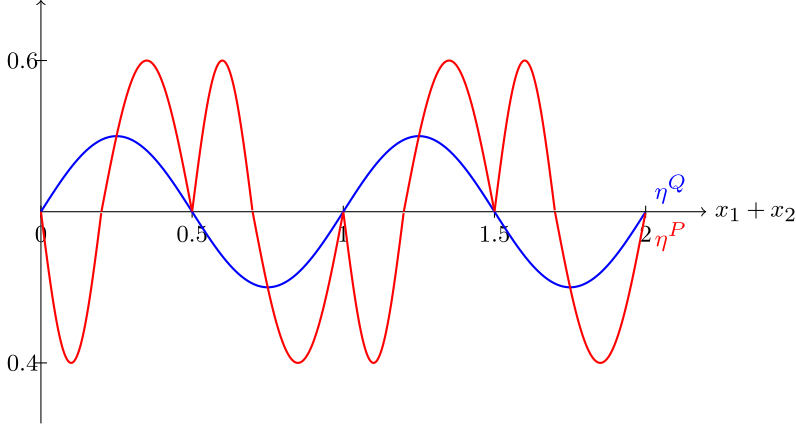


FIG. 1. Illustration of η^P in the second simulation setup with $\gamma = 1$ and $r = 0.4$.

where $\{a\} = a - \lfloor a \rfloor$ represents the fractional part of a real value a . The following graph illustrates our setup of η^P . While keeping η^P continuous with $\beta_P = 1$, the ratio parameter r creates an area where the Bayes classifier differs from the target distribution. The positive classification regimes are identical when $r = 0$ and completely opposite when $r = 1$. See Figure 1 for a visualization.

In each scenario, we set $n_Q = 200$ and $n_P = 1000$, as a large n_P is necessary to observe the benefits of transfer learning. Previous studies Cai and Wei (2021), Reeve, Cannings and Samworth (2021) have shown that accuracy improves with increasing n_P . We also generate 50000 independent test pairs from Q . For the TAB K -NN classifier, we choose $k_Q = \lfloor n_Q^{\frac{2\beta}{2\beta+d}} \rfloor = 31$, $k_P = \lfloor n_P^{\frac{2\gamma\beta}{2\gamma\beta+d}} \rfloor$, and $\tau = 0.05$. As mentioned before, $\beta = 1$.

We choose simple K -NN classifiers trained on Q -data and P -data as two benchmarks for comparison. Moreover, we add the K -NN classifier on the pooled data, combining both the target and source with the nearest-neighbor parameter k chosen by 5-fold cross-validation. We also consider the weighted K -NN classifiers proposed in Cai and Wei (2021) with their indicated optimal weighting scheme $w_Q + w_P = 1$, $w_P/w_Q = (n_Q + n_P^{\frac{2\beta+d}{2\gamma\beta+d}})^{\frac{(\gamma-1)\beta}{2\beta+d}}$.

Figure 2 shows that our TAB classifier is accurate when Δ is small, as does K -NN on P -data. Furthermore, our TAB classifier outperforms K -NN on P -data, K -NN on pooled data, and the weighted K -NN by a significant margin for large Δ , demonstrating its ability to avoid negative transfer when the source data is unreliable. The pooling and weighting algorithm benchmarks improve the classification when Δ is small; however, they tend to break down when Δ is large.

Additionally, Figure 3 shows that our TAB classifier improves the accuracy when the ambiguity level, depicted by r , is small. In addition, our TAB classifier significantly outperforms the other three benchmarks when the ambiguity level is too large to benefit from transfer learning, conserving the classification ability using only Q -data.

6.2. Logistic regression setting. We next consider the scenario where both η^Q and η^P follow the logistic models (20), and the parameters are estimated by (23). We set $n_Q = 200$, $n_P = 500$, $Q_X = P_X = N(0, I_d)$ and simulate 50000 test data points. The linear coefficient is given by

$$\beta_Q = (0.5 \cdot \mathbf{1}_s, \mathbf{0}_{d-s}), \quad \beta_P = \left(1.5 \cdot \mathbf{1}_s, \frac{\|\beta_Q\|}{\sqrt{d-s}} \tan \Delta \cdot \mathbf{1}_{d-s} \right),$$

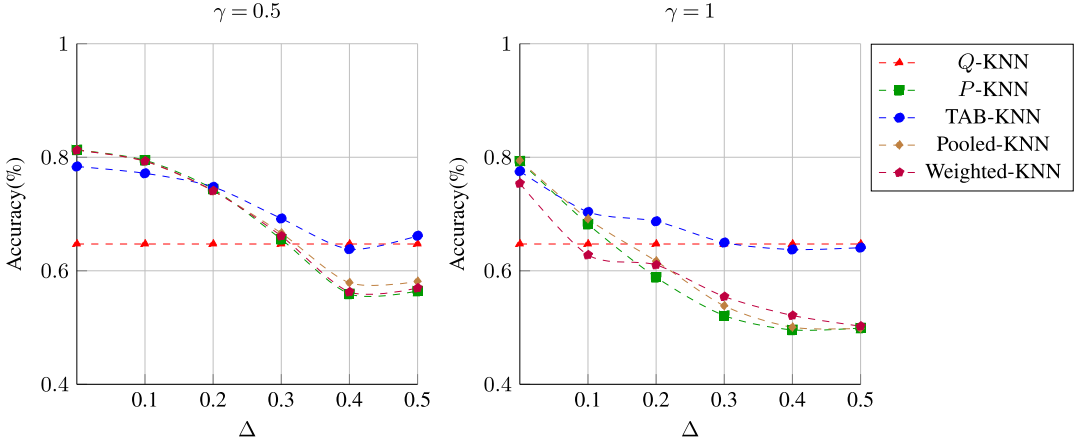


FIG. 2. Accuracy of the TAB K-NN classifiers under the band-like ambiguity scenario. We experiment with different values of Δ for a given $\gamma = 0.5$ and 1. Blue: TAB K-NN classifier; Red: K-NN classifier on only Q -data; Green: K-NN classifier on only P -data; Brown: K-NN classifier on pooled data.

where $\mathbf{1}_s$ and $\mathbf{0}_s$, respectively, denotes a vector of all 1s and a vector of all 0s with size s . Here, we set $s = 10$, which is small compared to d . For the source distribution, β_P could be treated as a rotated version of $3\beta_Q$, with an angle of exactly Δ between them. The range of Δ is chosen in the set $\{0, 0.2, 0.4, 0.6, 0.8, 1, 1.2, 1.4, 1.6\}$, which gradually approaches $\pi/2$. We simulate 50000 test data points for each value of Δ . For the TAB threshold, we also choose $\tau = 0.05$. The lasso regularization parameter is selected by 5-fold cross-validation and chosen to be the largest λ at which the MSE is within one standard error of the minimum MSE, namely λ_{1se} .

In addition to our proposed TAB classifier, we compare three benchmarks for performance: logistic regression with lasso penalty on Q -data, P -data and pooled data. Figure 4 demonstrates that our TAB classifier achieves high accuracy when the angle between β_Q and β_P is small. Interestingly, as evidence of its robustness, our classifier retains some classification ability with the target data even when the angle is large, while the benchmark classifiers based on P -data and pooled data suffer from negative transfer.

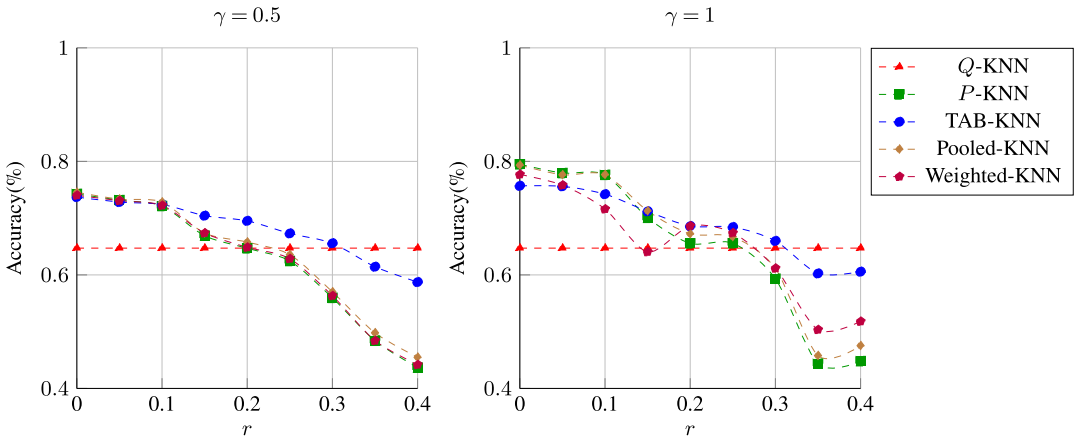


FIG. 3. Accuracy of the TAB K-NN classifiers under the scenario with partially flipped sine functions. We experiment with different values of the ratio parameter r for a given $\gamma = 0.5$ and 1. Blue: TAB K-NN classifier; Red: K-NN classifier on only Q -data; Green: K-NN classifier on only P -data; Brown: K-NN classifier on pooled data.

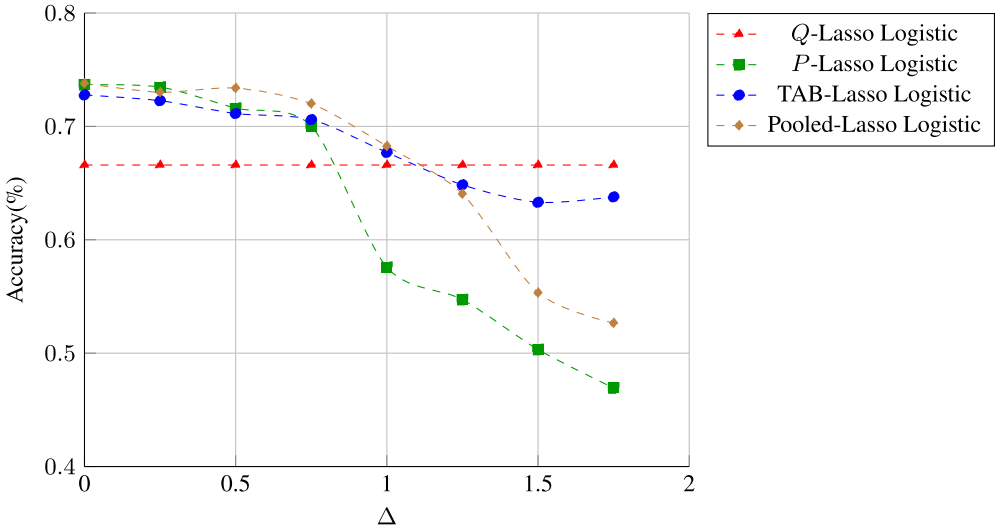


FIG. 4. Accuracy of the TAB logistic classifier with lasso penalty. We conduct experiments with different choices of the angle $\Delta \in [0, \pi/2]$. Blue: TAB logistic classifiers with lasso penalty; Red: Logistic classifier with lasso penalty on only Q -data; Green: Logistic classifier with lasso penalty on only P -data; Brown: Logistic classifier with lasso penalty on pooled data.

7. Conclusion. In this paper, we have proposed a new approach to transfer learning that is robust against an unreliable source distribution with arbitrary ambiguity in the source data. Our work uses a different way of transferring the source data and, in particular, encompasses the nonparametric setting in Cai and Wei (2021) and Reeve, Cannings and Samworth (2021) and parametric setting in Li, Cai and Li (2022). By introducing the ambiguity level, our approach enables us to understand the circumstances under which we can improve classification performance, given source data with potential ambiguity. Our proposed TAB classifier, with a threshold τ that balances the performance of both the target and source data, is shown to be both *efficient* and *robust*, as the excess risk improves for a reliable source distribution and avoids negative transfer with an unreliable source distribution. Furthermore, we provide simple approaches to bound the signal transfer risk, a key component of the excess risk in our general convergence result, in terms of the conventional excess risk extensively studied in the literature of statistical learning.

We then demonstrate the power of our approach on specific classification tasks, with a focus on nonparametric classification and logistic regression settings. The upper bounds are shown to be optimal up to some logarithmic factors and are more general than previous work on transfer learning. Simulation studies provide numerical evidence for these two classification tasks.

There are several promising avenues for future research that may build on the contributions of this paper. One potential direction is to consider an extension of the signal strength and ambiguity level that incorporates a translation parameter, that is,

$$s_\kappa(x) := \begin{cases} \left| \eta^P(x) - \frac{1}{2} - \kappa \right|, & \text{sgn}\left(\eta^Q(x) - \frac{1}{2} - \kappa\right) \times \left(\eta^P(x) - \frac{1}{2}\right) \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

$$\mathbb{E}_{(X,Y) \sim Q} \left[\left| \eta^Q(X) - \frac{1}{2} \right| \mathbf{1}_{\left\{ s_\kappa(X) \leq C_\gamma \left| \eta^Q(X) - \frac{1}{2} \right|^\gamma \leq C_\gamma z^\gamma \right\}} \right] \leq \varepsilon_\kappa(z; \gamma, C_\gamma).$$

This extension is natural since the decision boundary $\{x \in \Omega : \eta^Q(x) = 1/2\}$ may be similar to $\{x \in \Omega : \eta^P(x) = 1/2 + \kappa\}$ (they are identical under the settings of Reeve and Kaban

(2019) and Abramovich and Grinshtein (2019)). We conjecture that an additional estimation error of $(\frac{\log n_Q}{n_Q})^{\frac{1+\alpha}{2+\alpha}}$ may be incurred due to the presence of an unknown κ , and an empirical risk minimization procedure after obtaining $\hat{\eta}^P$ such as

$$\hat{\kappa} = \arg \min_{\kappa \in [-\frac{1}{2}, \frac{1}{2}]} \frac{1}{n} \sum_{i=1}^n \mathbf{1} \left\{ \mathbf{1} \left\{ \hat{\eta}^P(X_i) \geq \frac{1}{2} + \kappa \right\} \neq Y_i \right\}$$

may be required to obtain a corrected version of $\hat{f}^P$ over the target data.

Another direction is to develop an adaptive and generic procedure for selecting the threshold τ while preserving optimality. There are three possible directions. First, it is an open question whether simple Empirical Risk Minimization (ERM) methods for choosing τ still keep the upper bound optimal. Second, Lepski's method (Lepskiĭ (1992)) may offer a solution for maintaining optimal rates with an adaptive choice of τ . Third, we conjecture that choosing $\tau \asymp \log(n_Q \vee n_P) k_Q^{-1/2}$, where $\hat{\eta}^Q$ is an M -estimator consisting of a simple average of k_Q terms, is helpful for obtaining an optimal rate.

As a final future direction, the conditions presented in the nonparametric and logistic model settings could be relaxed to some extent, for instance, by considering noncompact feature spaces with sub-Gaussian conditions or other marginal distribution assumptions (e.g., Assumption A4 in Gadat, Klein and Marteau (2016)). Our proposed TAB classifier is expected to perform well in these settings without a significant modification of the framework, and may offer advantages over other case-specific estimators in the transfer learning literature.

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SUPPLEMENTARY MATERIAL

Supplement to “Robust transfer learning with unreliable source data” (DOI: [10.1214/25-AOS2534SUPP](https://doi.org/10.1214/25-AOS2534SUPP); .pdf). Please refer to the Supplementary Material for a more detailed explanation of certain topics and proofs of the results.

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